

Physics

1. If Surface tension (S), Moment of Inertia (I) and Planck's constant (h), were to be taken as the fundamental units, the dimensional formula for linear momentum would be:

A. $S^{3/2}I^{1/2}h^0$ B. $S^{1/2}I^{1/2}h^0$
 C. $S^{1/2}I^{3/2}h^{-1}$ D. $S^{1/2}I^{1/2}h^{-1}$

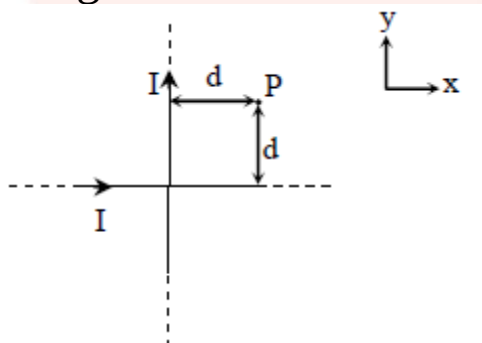
Ans. B

Explanation:

We know that

$$\begin{aligned} \text{linear momentum, } p &\propto \sqrt{\text{mass} \times \text{Energy}} \\ &\propto \sqrt{mR^2S} \\ &\propto \sqrt{IS} \\ [P] &= \left[I^{\frac{1}{2}} S^{\frac{1}{2}} \right] \end{aligned}$$

2. Two very long, straight, and insulated wires are kept at 90° angle from each other in xy-plane as shown in the figure.



These wires carry currents of equal magnitude I , whose directions are shown in the figure. The net magnetic field at point P will be:

- A. Zero B. $\frac{\mu_0 I}{2\pi d}(\hat{x} + \hat{y})$
 C. $-\frac{\mu_0 I}{2\pi d}(\hat{x} + \hat{y})$ D. $\frac{+\mu_0 I}{\pi d}(\hat{z})$

Ans. A

$$\vec{B} = B_1 + B_2 = \frac{\mu_0 I}{2\pi d} \hat{k} - \frac{\mu_0 I}{2\pi d} \hat{k}$$

$$\vec{B} = \vec{0}$$

3. A parallel plate capacitor has $1\mu\text{F}$ capacitance. One of its two plates is given $+2\mu\text{C}$ charge and the other plate, $+4\mu\text{C}$ charge. The potential difference developed across the capacitor is:

- A. 2V B. 3V
 C. 1V D. 5V

Ans. C

Given $C = 1\mu\text{F}$, $Q_1 = +4\mu\text{C}$, $Q_2 = +2\mu\text{C}$

$$q_{\text{cap}} = \frac{Q_1 - Q_2}{2} = \frac{4 - 2}{2} = 1\mu\text{C}$$

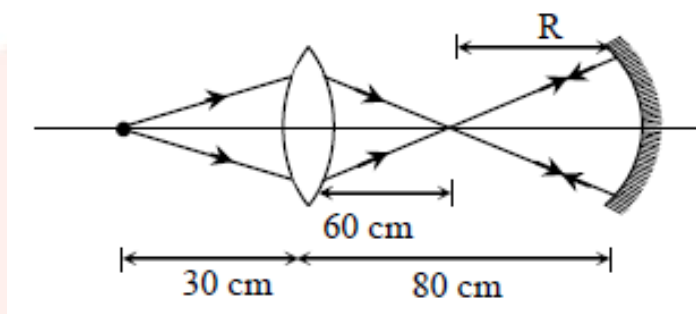
$$V = \frac{q}{C} = \frac{1 \times 10^{-6}}{1 \times 10^{-6}} = 1 \text{ volt}$$

4. A convex lens {of focal length 20cm) and a concave mirror, having their principal axes along the same lines, are kept 80cm apart from each other. The concave mirror is to the right of the convex lens. When an object is kept at a distance of 30cm to the left of the convex lens, its image remains at the same position even if the concave mirror is removed. The maximum distance of the object for which this concave mirror, by itself would produce a virtual image would be

- A. 30 cm B. 25 cm
C. 20 cm D. 10 cm

Ans. D

The given situation can be drawn as shown below



For lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{v} - \frac{1}{-30} = \frac{1}{20} \Rightarrow v = 60 \text{ cm}$$

$$R = 2f = 20 \text{ cm}$$

$$f = 10 \text{ cm}$$

Max distance for virtual image = $f = 10 \text{ cm}$

5. A body of mass m_1 moving with an unknown velocity of $v_1 \hat{i}$, undergoes a collinear collision with a body of mass m_2 moving with a velocity $v_2 \hat{i}$. After collision, m_1 and m_2 move with velocities of $v_3 \hat{i}$ and $v_4 \hat{i}$, respectively. If $m_3 = 0.5 m_1$ and $v_3 = 0.5 v_1$, then v_1 is

- A. $v_4 + v_2$ B. $v_4 - \frac{v_2}{4}$
C. $v_4 - \frac{v_2}{2}$ D. $v_4 - v_2$

Ans. D

Given,

$$m_2 = 0.5m_1 \Rightarrow m_1 = 2m_2$$

Let $m_2 = m$, then, $m_1 = 2m$

$$\text{Also, } v_3 = 0.5 v_1$$

$\therefore$ According to the conservation of linear momentum.

$$m_1 v_1 \hat{i} + m_2 v_2 \hat{i} = m_1 (0.5 v_1) \hat{i} + m_2 v_4 \hat{i}$$

$$m_1 v_1 \hat{i} + \frac{m_1 v_2}{2} = \frac{m_1 v_1}{2} \hat{i} + \frac{m_1 v_4}{2} \hat{i}$$

$$v_1 = v_4 = v_2$$

6. The electric field in a region is given by $\vec{E} = (Ax+B) \hat{i}$, where E is NC^{-1} and x is in meters. The values of constants are $A=20$ SI unit and $B=10$ SI unit. If the potential at $x=1$ is V_1 and that at $x=-5$ is V_2 , then V_1-V_2 is

- A. -48 V B. 320 V
C. -520 V D. 180 V

Ans. D

As we know that

$$dV = -E \cdot dr$$

$$dV = -Ax - B$$

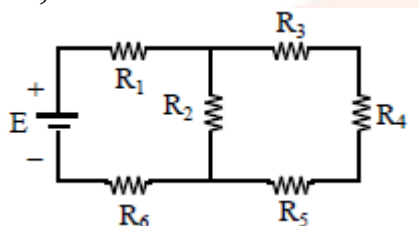
$$\int_{V_2}^{V_1} dV = \int_{-5}^1 -Ax - B$$

$$\Delta V = -\frac{A}{2} [x^2]_{-5}^1 - B[x]_{-5}^1$$

$$= -10(24) - 60$$

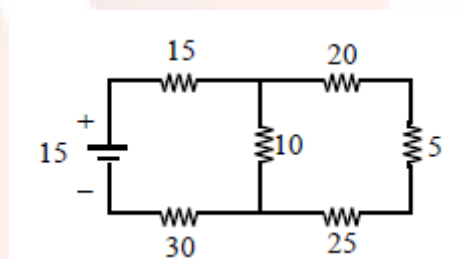
$$= 180$$

7. In the figure shown, what is the current (in Ampere) drawn from the battery? You are given:
 $R_1=15\ \Omega$, $R_2=10\ \Omega$, $R_3=20\ \Omega$, $R_4=5\ \Omega$, $R_5=25\ \Omega$, $R_6=30\ \Omega$, $E=15V$



- A. $13/24$ B. $7/18$
 C. $9/32$ D. $20/3$

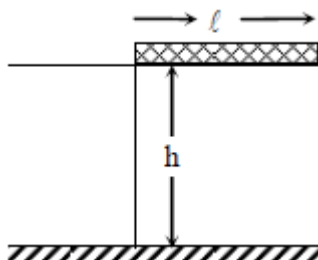
Ans. C



$$i = \frac{\text{Voltage}}{\text{Net resistance of the circuit}}$$

$$i = \frac{15}{45 + \frac{25}{3}} = \frac{3}{9 + \frac{5}{3}} = \frac{9}{32}$$

8. A rectangular solid box of length 0.3 is held horizontally, with one of its sides on the edge of a platform of height 5 m. When released, it slips off the table in a very short time $t=0.01$ s, remaining essentially horizontal. The angle by which it would rotate when it hits the ground will be (in radians) close to



- A. 0.5 B. 0.02
C. 0.3 D. 0.28

Ans. A

Initial angular acceleration before it slips off

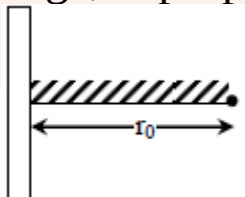
$$\alpha = \frac{MgL / 2}{ML^2 / 3} = \frac{3g}{2L}$$

$$\omega = \frac{3g}{2L} \tau = \frac{30(0.01)}{2(0.3)} = 0.5$$

$$t = \sqrt{\frac{2 \times 5}{10}} = 1 \text{ sec}$$

$$\Delta\theta = \omega t = (0.5)(1) = 0.5 \text{ Rad}$$

9. A positive point charge is released from rest at a distance r_0 from a positive line charge with uniform density. The speed (v) of the point charge, as a function of instantaneous distance r from line charge, is proportional to:



A. $v \propto e^{+r/r_0}$ B. $v \propto \left(\frac{r}{r_0}\right)$
 C. $v \propto \sqrt{\ln\left(\frac{r}{r_0}\right)}$ D. $v \propto \ln\left(\frac{r}{r_0}\right)$

Ans. C

For a positive line charge or charged wire with uniform density λ , electric field at distance x is

$$E = \frac{2k\lambda}{r} = \frac{\lambda}{2\pi\epsilon_0 r}$$

Total work done by field of wire in taking charge q from distance r_0 to distance r will be

$$= 2kq\lambda \log \left| \frac{r}{r_0} \right|$$

$$\frac{1}{2}mv^2 \propto \ln\left(\frac{r}{r_0}\right)$$

$$v \propto \sqrt{\ln\left(\frac{r}{r_0}\right)}$$

- 10.** A damped harmonic oscillator has a frequency of 5 oscillations per second. The amplitude drops to half its value for every 10 oscillations. The time it will take to drop to $\frac{1}{1000}$ of the original amplitude is close to:

A. 50s B. 10s
 C. 20s D. 100s

Ans. C

Given, frequency of oscillations is $f = 5 \text{ osc s}^{-1}$

$\Rightarrow$ Time period of oscillations is $T = \frac{1}{f} = \frac{1}{5} \text{ s}$

So, time for 10 oscillations is $= \frac{10}{5} = 2 \text{ s}$

$$(2)^{10} \approx 1024$$

So time $t = 10 T$
 $= 20 \text{ sec}$

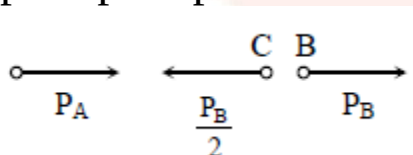
- 11.** A nucleus A, with a finite de-broglie wavelength λ_A , undergoes spontaneous fission into two nuclei B and C of equal mass. B flies in the same direction as that of A, while C flies in the opposite direction with a velocity equal to half of that of B. The de-Broglie wavelengths λ_B and λ_C of B and C are respectively:

- A. $\frac{\lambda_A}{2}, \lambda_A$ B. $\lambda_A, 2\lambda_A$
 C. $\lambda_A, \frac{\lambda_A}{2}$ D. $2\lambda_A, \lambda_A$

Ans. A

According to the conservation of linear momentum,
 Initial momentum = Final momentum

$$p_A = p_B + p_C$$



$$\lambda_B = \frac{\lambda_A}{2}; \quad \lambda_C = 2\lambda_B = \lambda_A$$

$$P_A = P_B - \frac{P_B}{2} \Rightarrow P_B = 2P_A$$

- 12.** A solid sphere and solid cylinder of identical radii approach an incline with the same linear velocity (see figure). Both roll without slipping all throughout. The two climb maximum heights h_{sph} and h_{cyl} on the incline. The ratio $\frac{h_{\text{sph}}}{h_{\text{cyl}}}$ is given by:



- A. $\frac{4}{5}$ B. $\frac{2}{\sqrt{5}}$
C. 1 D. $\frac{14}{15}$

Ans. D

Now from the conservation of mechanical energy

$$K_{\text{total}} = mgh$$

$$\frac{1}{2}mv^2(1 + \eta) = mgh$$

$$h \propto 1 + \eta$$

$$\frac{h_{\text{sp}}}{h_{\text{cyl}}} = \frac{1 + \frac{2}{5}}{1 + \frac{1}{2}} = \frac{\frac{7}{5}}{\frac{3}{2}} = \frac{14}{15}$$

13. The ratio of mass densities of nuclei of ^{40}Ca and ^{16}O is close to:

- A. 5 B. 2
C. 0.1 D. 1

Ans. D

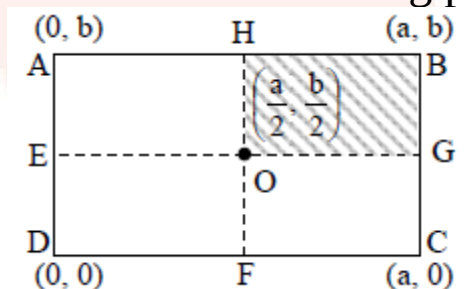
Mass density of nuclear matter is a constant quantity for all elements. It does not depend on elements mass number or atomic radius.

$$\rho_{\text{nucleus}} \propto A^0$$

$$\frac{\rho_{\text{Ca}}}{\rho_{\text{O}}} = \frac{1}{1}$$

$$1 : 1$$

14. A uniform rectangular thin sheet ABCD of mass M has length a and breadth b, as shown in the figure. If the shaded portion HBGO is cut-off, the coordinates of the center of mass of the remaining portion will be:



- A. $\left(\frac{5a}{12}, \frac{5b}{12}\right)$ B. $\left(\frac{3a}{4}, \frac{3b}{4}\right)$
C. $\left(\frac{5a}{3}, \frac{5b}{3}\right)$ D. $\left(\frac{2a}{3}, \frac{2b}{3}\right)$

Ans. A

Using formula for centre of mass, we have

$$X_{CM} = \frac{A_1 x_1 - A_2 x_2}{A_1 - A_2}$$

$$X_c = \frac{2m\left(\frac{a}{4}\right) + m\left(\frac{3a}{4}\right)}{3m} = \frac{5a}{12}$$

$$Y_c = \frac{2m\left(\frac{b}{2}\right) + m\left(\frac{b}{4}\right)}{3m} = \frac{5b}{12}$$

- 15.** Calculate the limit of resolution of a telescope objective having a diameter of 200 cm, if it has to detect light of wavelength 500 nm coming from a star

- A. 610×10^{-9} radian
- B. 152.5×10^{-9} radian
- C. 305×10^{-9} radian
- D. 457.5×10^{-9} radian

Ans. B

For a telescope, limit of resolution is given by $\Delta\theta = \frac{1.22\lambda}{D}$

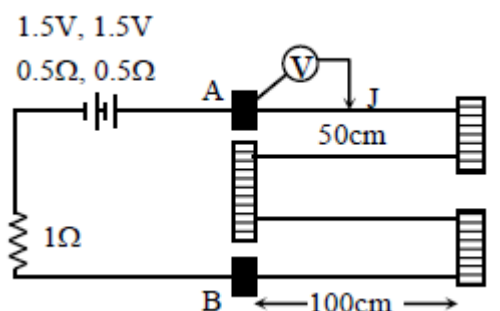
Here, $\lambda = 500 \text{ nm} = 500 \times 10^{-9} \text{ m}$

$D = 200 \text{ cm} = 200 \times 10^{-2} \text{ m}$

$$\Delta\theta = \frac{1.22\lambda}{d} = \frac{1.22 \times 500 \times 10^{-9}}{200 \times 10^{-2}} = 305 \times 10^{-9} \text{ rad}$$

- 16.** In the circuit shown, a four-wire potentiometer is made of a 400 cm long wire, which extends between A and B. The resistance per unit length of the potentiometer wire is $r=0.01 \Omega/\text{cm}$. If an ideal voltmeter is connected as shown with

jockey J at 50 cm from end A, the expected reading of the voltmeter will be :



- A. 0.20 V B. 0.25 V
C. 0.75 V D. 0.50 V

Ans. B

In given potentiometer, resistance per unit length is $x = 0.01 \Omega \text{ cm}^{-1}$

Length of potentiometer wire is $L = 400 \text{ cm}$.

$\therefore$ Current in given potentiometer circuit is

$$i = \frac{E}{R + r + I_R}$$

$$i = \frac{3}{1 + 1 + 4} = 0.5 \text{ A}$$

$$V = KL = 0.5 \times 0.5 = 0.25 \text{ volt}$$

17. The temperature, at which the root mean square velocity of hydrogen molecules equals their escape velocity from the earth, is closest to : [Boltzmann Constant $K_B = 138 \times 10^{-23} \text{ J/K}$ Avogadro Number $N_A = 6.02 \times 10^{26} / \text{kg}$ Radius of Earth : $6.4 \times 10^6 \text{ m}$ Gravitational acceleration on Earth = 10 ms^{-2}]
A. 10^4 K B. 800 K

C. 650 K D. 3×10^5 K

Ans. A

As we know that

Root mean square velocity of hydrogen molecule is given as

$$v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}$$

Escape velocity of hydrogen molecule from the earth is given as

$$v_e = \sqrt{2gR_e}$$

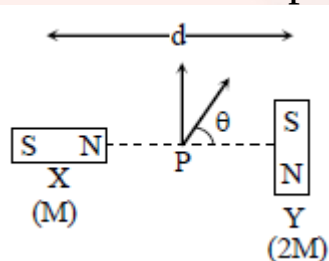
Given $v_{\text{rms}} = v_e$

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{2GM_E}{R_E}}$$

$$\frac{3 \times 8.314 \times T}{2 \times 10^{-3}} = (11.2 \times 10^3)^2$$

$$T = 10^4 \text{ K}$$

- 18.** Two magnetic dipoles X and Y are placed at a separation d , with their axes perpendicular to each other. The dipole moment of Y is twice that of X. A particle of charge q is passing through their mid-point P, at angle $\theta = 45^\circ$ with the horizontal line, as shown in figure. What would be the magnitude of force on the particle at that instant? (d is much larger than the dimensions of the dipole)



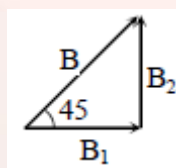
A. 0

B. $\sqrt{2} \left(\frac{\mu_0}{4\pi} \right) \frac{M}{\left(\frac{d}{2} \right)^3} \times qv$

C. $\left(\frac{\mu_0}{4\pi} \right) \frac{M}{\left(\frac{d}{2} \right)^3} \times qv$

D. $\left(\frac{\mu_0}{4\pi} \right) \frac{2M}{\left(\frac{d}{2} \right)^3} \times qv$

Ans. A



From the figure we can see that,

$$B_1 = \frac{\mu_0}{4\pi} \frac{2(M)}{(d/2)^3}$$

$$B_2 = \frac{\mu_0}{4\pi} \frac{(2M)}{(d/2)^3}$$

$$B_1 = B_2$$

Since, $\vec{v}$ is along $\vec{B}$

Hence, the magnitude of force on the particle at that instant will be $\vec{F}_{\text{net}} = \vec{0}$

19. In a line of sight radio communication, a distance of about 50 km is kept between the transmitting and receiving

antennas. If the height of the receiving antenna is 70m, then the minimum height of the transmitting antenna should be: (Radius of the Earth = 6.4×10^6 m).

- A. 51 m B. 20 m
C. 40 m D. 32 m

Ans. D

The distance is given

$$D = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

$$50 \times 10^3 = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

$$h_T = \left[5 \times 10^4 - \sqrt{2 \times 6.4 \times 10^6 \times 70} \right]^2$$

$$h_T \approx 32 \text{ m}$$

20. A circuit connected to an ac source of emf $e = e_0 \sin(100t)$ with t in seconds, gives a phase deference of $\frac{\pi}{4}$ between the emf e and current i . Which of the following circuits will exhibit this?

- A. RC circuit with $R = 1 \text{ k}\Omega$ and $C = 10 \mu\text{F}$
B. RC circuit with $R = 1 \text{ k}\Omega$ and $C = 1 \mu\text{F}$
C. RC circuit with $R = 1 \text{ k}\Omega$ and $L = 10 \text{ mH}$
D. RC circuit with $R = 1 \text{ k}\Omega$ and $L = 1 \text{ mH}$

Ans. A

$$\text{Since } \tan \frac{\pi}{4} = 1 = \frac{X_C \text{ or } X_L}{R}$$

$\therefore$ For R-C circuit, we have

$$1 = \frac{1}{C\omega R} \text{ or } \omega = \frac{1}{CR} \quad \dots(i)$$

Similarly, for R-L circuit, we have

$$1 = \frac{\omega L}{R} \Rightarrow \omega = \frac{R}{L} \quad \dots(ii)$$

It is given in the question that, $\omega = 100 \text{ rad/s}$. Thus, again by substituting the given values of R, C or L option wise in the respective Eqs. (i) and (ii), we get that only for option (a)

$$\omega = \frac{1}{CR} = \frac{1}{10 \times 10^{-6} \times 10^3}$$

or $\omega = 100 \text{ rad/s}$

21. Let $|\vec{A}_1| = 3$, $|\vec{A}_2| = 5$ and $|\vec{A}_1 + \vec{A}_2| = 5$. The value of

$(2\vec{A}_1 + 3\vec{A}_2) \cdot (3\vec{A}_1 + 2\vec{A}_2)$ is:

- A. -106.5 B. -99.5
C. -118.5 D. -112.5

Ans. C

As we know that $|\vec{A}_1 + \vec{A}_2|$ can also be written as

$$|\vec{A}_1 + \vec{A}_2| = \sqrt{|\vec{A}_1|^2 + |\vec{A}_2|^2 + 2|\vec{A}_1||\vec{A}_2|\cos\theta}$$

Substituting the given values, we get

$$5 = \sqrt{(3)^2 + (5)^2 + 2 \times 3 \times 5 \cos\theta}$$

$$\text{or } \cos\theta = -\frac{9}{2 \times 3 \times 5} = -\frac{3}{10}$$

$$(2\vec{A}_1 + 3\vec{A}_2) \cdot (3\vec{A}_1 + 2\vec{A}_2)$$

$$= 6A_1^2 - 4\vec{A}_1 \cdot \vec{A}_2 + 9\vec{A}_1 \cdot \vec{A}_2 - 6A_2^2$$

$$= 6(A_1^2 - A_2^2) + 5A_1A_2 \cos\theta$$

$$= -96 + 5A_1A_2 \cos\theta$$

$$25 = 9 + 25 + 2(15 \cos\theta)$$

$$A_1A_2 \cos\theta = -0/2$$

$$= -96 - \frac{5 \times 9}{2} = -118.5$$

22. The magnetic field of an electromagnetic wave is given by:

$$\vec{B} = 1.6 \times 10^{-6} \cos(2 \times 10^7 z + 6 \times 10^{15} t) (2\hat{i} + \hat{j}) \frac{\text{Wb}}{\text{m}^2}$$

The associated electric field will be:

A. $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t) (-\hat{i} + 2\hat{j}) \frac{\text{V}}{\text{m}}$

B. $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t) (-2\hat{j} + \hat{i}) \frac{\text{V}}{\text{m}}$

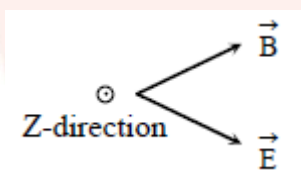
C. $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t) (2\hat{i} + \hat{j}) \frac{\text{V}}{\text{m}}$

D. $\vec{E} = 4.8 \times 10^2 \cos(2 \times 10^7 z + 6 \times 10^{15} t) (\hat{i} - 2\hat{j}) \frac{\text{V}}{\text{m}}$

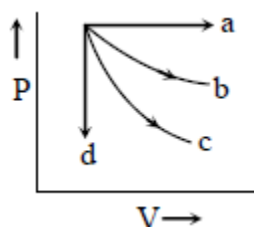
Ans. D

As we know that

$$\begin{aligned} E_0 &= CB_0 \\ &= 3 \times 10^8 \times 1.6 \times 10^{-6} \\ &= 4.8 \times 10^2 \end{aligned}$$

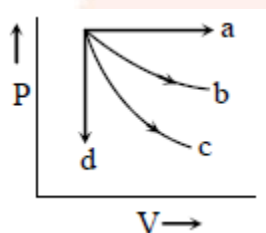


23. The given diagram shows four processes i.e., isochoric, isobaric, isothermal and adiabatic. The correct assignment of the processes, in the same order is given by:



- A. a d b c B. d a c b
C. a d c b D. d a b c

Ans. B



- a $\rightarrow$ isobaric
d $\rightarrow$ isochoric
b $\rightarrow$ isothermal
c $\rightarrow$ adiabatic
(d a b c)

- 24.** A rocket has to be launched from earth in such a way that it never returns. If E is the minimum energy delivered by the rocket launcher, what should be the minimum energy that the launcher should have if the same rocket is to be launched from the surface of the moon? Assume that the density of the earth and the moon are equal and that the earth's volume is 64 times the volume of the moon.

- A. $\frac{E}{4}$ B. $\frac{E}{32}$
C. $\frac{E}{16}$ D. $\frac{E}{64}$

Ans. C

escape energy of a planet is given by

$$E \propto R^2$$

the ratio of escape energy is

$$\frac{E'}{E} = \frac{R^2}{16R^2}$$

$$E' = \frac{E}{16}$$

25. An electric dipole is formed by two equal and opposite charges q with separation d . The charges have same mass m . It is kept in a uniform electric field E . If it is slightly rotated from its equilibrium orientation, then its angular frequency ω is:

- A. $\sqrt{\frac{qE}{md}}$ B. $\sqrt{\frac{qE}{2md}}$
C. $\sqrt{\frac{2qE}{md}}$ D. $2\sqrt{\frac{qE}{md}}$

Ans. B

The general equation of angular SHM i.e.,

$$\alpha = -\omega^2 \theta$$

where, ω is the angular frequency we get

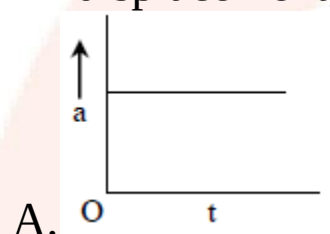
$$\omega^2 = \frac{2pE}{md^2} \text{ or } \omega = \sqrt{\frac{2pE}{md^2}}$$

As, $p = qd$

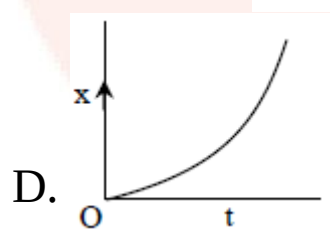
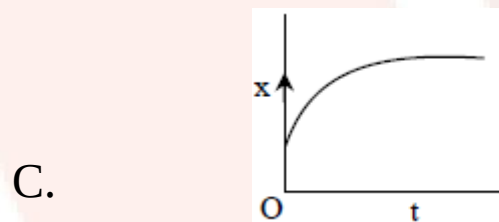
$$\therefore \omega = \sqrt{\frac{2qdE}{md^2}} = \sqrt{\frac{2qE}{md}}$$

26. A particle starts from origin O from rest and moves with a uniform acceleration along the positive x-axis. Identify all figures that correctly represent the motion qualitatively.

(a = acceleration, v = velocity,
x = displacement, t = time)



B.

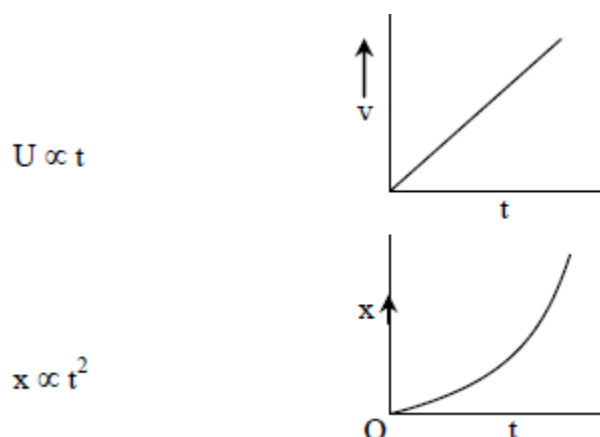


A. B, C B. A, B, C

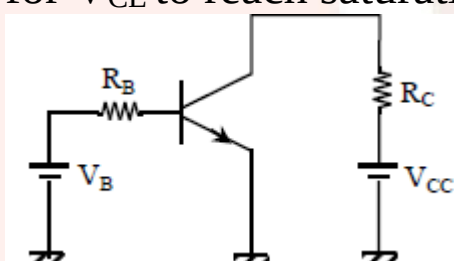
C. A D. A, B, D

Ans. D

a = constant



27. A common emitter amplifier circuit, built using an npn transistor, is shown in the figure. Its dc current gain is 250, $R_C = 1 \text{ k}\Omega$ and $V_{CC} = 10 \text{ V}$. What is the minimum base current for V_{CE} to reach saturation?



- A. $10 \mu\text{A}$ B. $7 \mu\text{A}$
C. $100 \mu\text{A}$ D. $40 \mu\text{A}$

Ans. D

For a common emitter n-p-n transistor, DC current gain is

$$\beta_{DC} = \frac{I_C}{I_B}$$

For saturation, $V_{CC} - I_C R_C = 0$

$$10 - I_C \times 10^3 = 0$$

$$I_C = 10^{-2}$$

$$I_B = \frac{I_C}{\beta} = \frac{10^{-2}}{250} = 40 \mu\text{A}$$

28. A cell of internal resistance r drives current through an external resistance R . The power delivered by the cell to the external resistance will be maximum when:

- A. $R=r$ B. $R=1000\ r$
C. $R=0.001\ r$ D. $R=2r$

Ans. A

For maximum power in external resistance,

Internal resistance = External resistance

$$R = r$$

29. Young's moduli of two wires A and B are in the ratio 7:4. Wire A is 2m long and has radius R . Wire B is 1.5 m long and has radius 2 mm. If the two wires stretch by the same length for a given load, then the value of R is close to:

- A. 1.9 mm B. 1.5 mm
C. 1.3 mm D. 1.7 mm

Ans. D

When a wire is stretched then change in length of wire is

$$\frac{F}{A} = Y \frac{\Delta \ell}{\ell}$$

$$F \propto \frac{AY}{\ell}$$

$$\frac{A_1 Y_1}{\ell_1} = \frac{A_2 Y_2}{\ell_2}$$

$$\frac{R^2(7)}{2} = \frac{2^2(4)}{1.5}$$

$$R = 1.74$$

30. In a simple pendulum experiment for determination of acceleration due to gravity (g), time taken for 20 oscillations is measured by using a watch of 1 second least count. The mean value of time taken comes out to be 30 s. The length of pendulum is measured by using a meter scale of least count 1 mm and the value obtained is 55.0 cm. The percentage error in the determination of g is close to:

- A. 3.5% B. 0.7%
C. 0.2% D. 6.8%

Ans. D

Relation used for finding acceleration due to gravity by using a pendulum is

$$g = \frac{4\pi^2 l}{T^2}$$

So, fractional error in value of g is

$$\frac{\Delta g}{g} = \frac{\Delta l}{l} + \frac{2\Delta T}{T}$$

Given, $\Delta l = 0.1 \text{ cm}$, $l = 55 \text{ cm}$, $\Delta T = 1 \text{ s}$ and T for 20 oscillations = 30 s

$$\% \left| \frac{\Delta g}{g} \right| = \frac{0.1}{55} \times 100 + 2 \frac{1}{30} \times 100$$

$$= 6.8 \%$$

Chemistry

1. Which of the following compounds will show the maximum 'enol' content?

- A. $\text{CH}_3\text{COCH}_2\text{CONH}_2$
- B. $\text{CH}_3\text{COCH}_2\text{COOC}_2\text{H}_5$
- C. $\text{CH}_3\text{COCH}_2\text{COCH}_3$
- D. CH_3COCH_3

Ans. B

2. 5 moles of an ideal gas at 100K are allowed to undergo reversible compression till its temperature becomes 200 K. If $C_V = 28 \text{ K}^{-1} \text{ mol}^{-1}$, calculate ΔU and ΔpV for this process.

($R = 8.0 \text{ J K}^{-1} \text{ mol}^{-1}$)

- A. $\Delta U = 14 \text{ KJ}$; $\Delta(pV) = 18 \text{ kJ}$
- B. $\Delta U = 2.8 \text{ KJ}$; $\Delta(pV) = 0.8 \text{ kJ}$
- C. $\Delta U = 14 \text{ KJ}$; $\Delta(pV) = 4 \text{ kJ}$
- D. $\Delta U = 14 \text{ j}$; $\Delta(pV) = 0.8 \text{ kJ}$

Ans. C

Given that

$n = 5 \text{ mole}$, $T_2 = 200 \text{ K}$, $T_1 = 100\text{K}$

$C_V = 28 \text{ JK}^{-1} \text{ mol}^{-1}$.

As we know that

$$\Delta U = n C_V \Delta T,$$

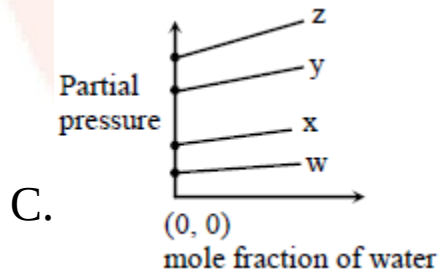
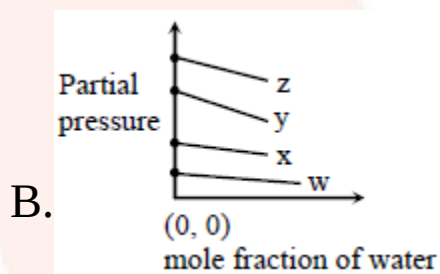
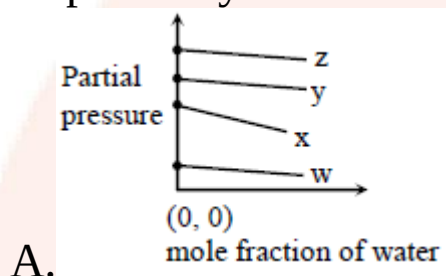
$$= 5(28) (100) \text{ J}$$

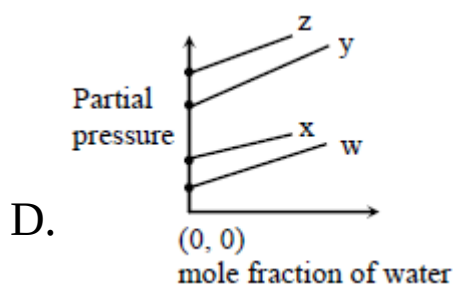
$$= 14000 \text{ J} = 14 \text{ kJ}$$

$$\Delta PV = P_2 V_2 - P_1 V_1$$

$$\begin{aligned} & nRT_2 - nRT_1 \\ & = nR (T_2 - T_1) \\ & = 5(8) (100) = 4000 \text{ J} = 4 \text{ kJ} \end{aligned}$$

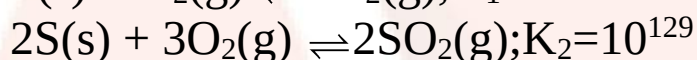
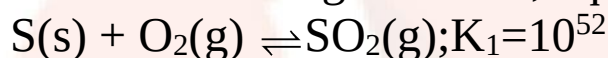
3. For the solution of the gases w, x, y and z in water at 293 K, the Henry's law constants (K_H) are 0.5, 2, 35 and 40 k bar, respectively. The correct plot for the given data is:



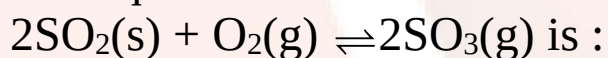


Ans. B

4. For the following reactions, equilibrium constants are given:



The equilibrium constant for the reaction,

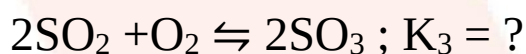
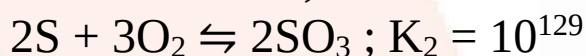
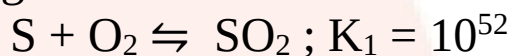


A. 10^{77} B. 10^{154}

C. 10^{25} D. 10^{181}

Ans. C

The given reactions



The value of equilibrium constant,

$$K_3 = K_1^{-2} \cdot K_2 = \frac{K_2}{K_1^2} = \frac{10^{129}}{10^{104}} = 10^{25}$$

5. If p is the momentum of the fastest electron ejected from a metal surface after the irradiation of light having wavelength λ , then for $1.5 p$ momentum of the photoelectron, the wavelength of the light should be: (Assume kinetic energy of

ejected photoelectron to be very high in comparison to work function):

- A. $\frac{1}{2}\lambda$ B. $\frac{4}{9}\lambda$
C. $\frac{3}{4}\lambda$ D. $\frac{2}{3}\lambda$

Ans. B

In photoelectric effect,

$$E = w + KE$$

It is given that KE of ejected electron is very high in comparison to work function w .

$$E = KE$$

$$\frac{hc}{\lambda} = \frac{p^2}{2m}$$

$$p^2 \propto \frac{1}{\lambda}$$

$$\left(\frac{p_2}{p_1}\right)^2 = \frac{\lambda_1}{\lambda_2}$$

$$\left(\frac{1.5p_1}{p_1}\right)^2 = \frac{\lambda_1}{\lambda_2}$$

$$\left(\frac{3}{2}\right)^2 = \frac{\lambda_1}{\lambda_2}$$

$$\frac{9}{4} = \frac{\lambda_1}{\lambda_2}$$

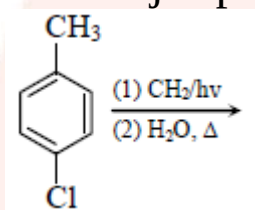
$$\lambda_2 = \frac{4}{9}\lambda_1$$

6. The statement that is INCORRECT about the interstitial compounds is:

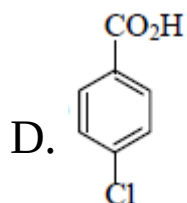
- A. they are very hard.
- B. they are chemically reactive.
- C. they have high melting points.
- D. they have metallic conductivity.

Ans. B

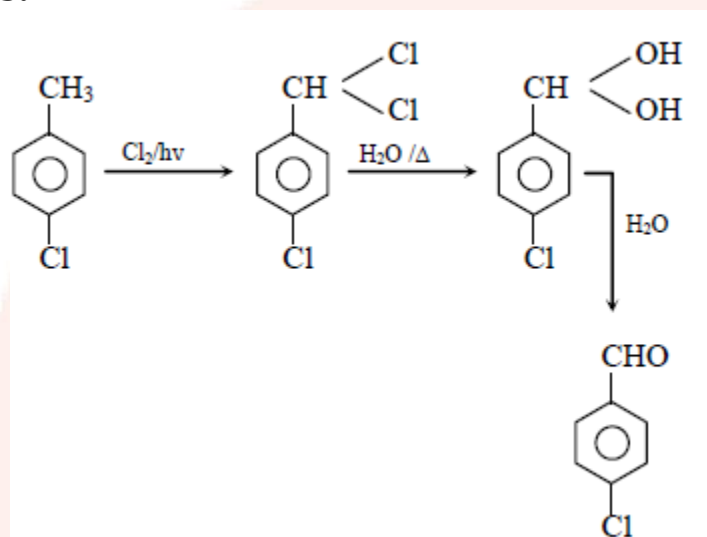
7. The major product of the following reaction is:



- A.
- B.
- C.



Ans. A



8. The calculated spin-only magnetic moments (BM) of the anionic and cationic species of $[\text{Fe}(\text{H}_2\text{O})_6]_2$ and $[\text{Fe}(\text{CN})_6]$, respectively, are:

A. 0 and 5.92 B. 4.9 and 0
C. 2.84 and 5.92 D. 0 and 4.9

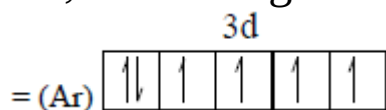
Ans. B

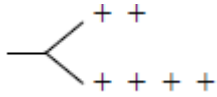
The given Compound is $[\text{Fe}(\text{H}_2\text{O})_6]_2 [\text{Fe}(\text{CN})_6]$

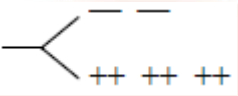
The Cation in the given compound is $[\text{Fe}(\text{H}_2\text{O})_6]^{+2}$

The Anion in the given compound is $[\text{Fe}(\text{CN})_6]^{-4}$

So, the Configuration of $\text{Fe}^{+2} = (\text{Ar}) 3d^6$



For H_2O W.F. ligand 
 4 unpaired e^- , $\therefore \mu \sqrt{4(4+2)} = 4.9 = 4.9 \text{ B.M.}$

For CN^- & F^- ligand 
 No unpaired e^-
 $\mu = 0$

9. The covalent alkaline earth metal halide ($\text{X} = \text{Cl}, \text{Br}, \text{I}$) is :

- A. MgX_2 B. BeX_2
 C. SrX_2 D. CaX_2

Ans. B

According to Fajan's rule, greater the polarizing power of cation greater would be the covalent character. Since Be^{2+} has maximum polarizing power among given cation, therefore, BeX_2 would be most covalent among given alkaline with metal halides.

10. 0.27 g of a long chain fatty acid was dissolved in 100 m^3 of hexane. 10 mL of this solution was added drop wise to the surface of water in a round watch glass. Hexane evaporates and a monolayer is formed. The distance from edge to center of the watch glass is 10 cm. What is the height of the monolayer? [Density of fatty acid = 0.9 g cm^{-3} ; $\pi = 3$]
 A. 10^{-8} m B. 10^{-6} m
 C. 10^{-4} m D. 10^{-2} m

Ans. B

100 mL (cm^3) of hexane contains 0.27 g of fatty acid.

In 10 mL solution, mass of the fatty acid.

$$m = \frac{0.27}{100} \times 10 = 0.027 \text{ g}$$

In 100ml gm of fatty acid = 0.27 gm

$$1 \text{ ml gm} \dots\dots\dots = \frac{0.27}{100}$$

$$10 \text{ ml gm} \dots\dots\dots = \frac{0.27}{100} \times 10 = 0.027$$

$$d = \frac{m}{v}$$

$$d \times v = m$$

$$0.9 \left(\frac{\text{gm}}{\text{cm}^3} \right) \times \text{area} \times \text{height} = 0.027 \text{ gm}$$

$$0.9 \times (3) \times (10)^2 \times h = 0.027$$

$$h = 10^{-4} \text{ cm}$$

$$h = 10^{-6} \text{ m}$$

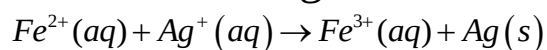
11. Polysubstitution is a major drawback in:

- A. Friedel Craft's alkylation
- B. Acetylation of aniline
- C. Friedel Craft's acylation
- D. Reimer Tiemann reaction

Ans. C

Polysubstitution is a major drawback of Friedal –Craft alkylation.

12. Calculate the standard cell potential (in V) of the cell in which following reaction takes place:



Given that

$$E_{Ag^+/Ag}^0 = xV$$

$$E_{Fe^{2+}/Fe}^0 = yV$$

$$E_{Fe^{3+}/Fe}^0 = zV$$

A. $x-y$ B. $x+2y-3z$

C. $x-y$ D. $x+y-z$

Ans. B

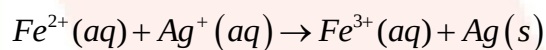
It is given that

$$E_{Ag^+/Ag}^0 = x$$

$$E_{Fe^{2+}/Fe}^0 = z$$

$$E_{Fe^{3+}/Fe}^0 = y$$

Now, for the given reaction:



$$E_{cell}^0 = \frac{E_C^0}{R.P.} - \frac{E_A^0}{R.P.}$$

$$= E_{Ag^+/Ag}^0 = E_{Fe^{3+}/Fe^{2+}}^0$$

$$= x - (3z - 2y)$$

$$= x + 2y - 3z$$

$$E_3 = \frac{\pm n_1, E_1 \pm n_2 E_2}{n_3}$$

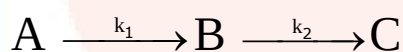
$$= \frac{3z - 2y}{1} = 3z - 2y$$

13. For a reaction scheme $A \xrightarrow{k_1} B \xrightarrow{k_2} C$, if the rate of formation of B is set to be zero then the concentration of B is given by:

- A. $k_1 k_2 [A]$
- B. $(k_1 - k_2) [A]$
- C. $(k_1 + k_2) [A]$
- D. $\left(\frac{k_1}{k_2}\right) [A]$

Ans. D

Given that



The rate of formation of B is

$$\frac{dB}{dt} = K_1 [A] - K_2 [B] = 0$$

$$K_1 [A] = K_2 [B]$$

Concentration of B is

$$[B] = \frac{K_1}{K_2} [A]$$

$$\frac{d_A}{dt} = K_1 [A]$$

$$\frac{d_C}{dt} = K_2 [B]$$

14. The correct statement about ICl_5 and ICl_4^- is:

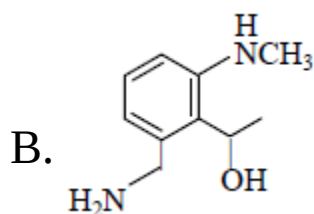
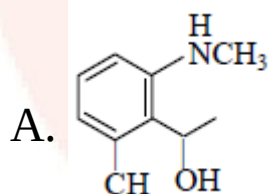
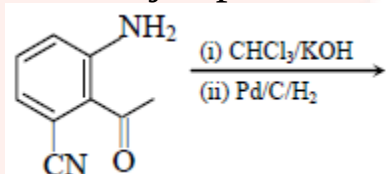
- A. ICl_5 is square pyramidal and ICl_4^- is square planar
- B. ICl_5 is square pyramidal and ICl_4^- is tetrahedral
- C. both are isostructural.
- D. ICl_5 is trigonal bipyramidal and ICl_4^- is tetrahedral

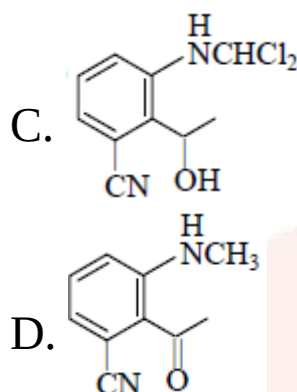
Ans. A

ICl_5 $\sigma = 5$ lp = 1 sp^3d^2 square pyramidal

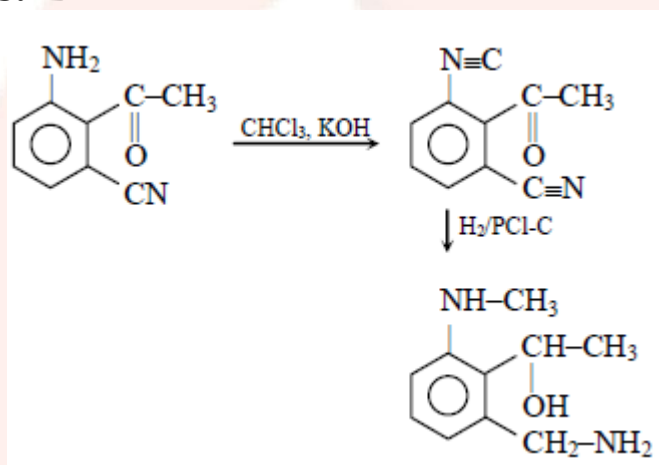
ICl_4^- $\sigma = 4$ lp = 2 sp^3d^2 square planar

15. The major product obtained in the following reaction is:





Ans. B



16. Fructose and glucose can be distinguished by:

- A. Benedict's test B. Fehling's test
C. Barfoed's test D. Seliwanoff's test

Ans. D

Seliwanoff's test is used to differentiate between ketose and aldose. Glucose and fructose can be distinguished by Seliwanoff's.

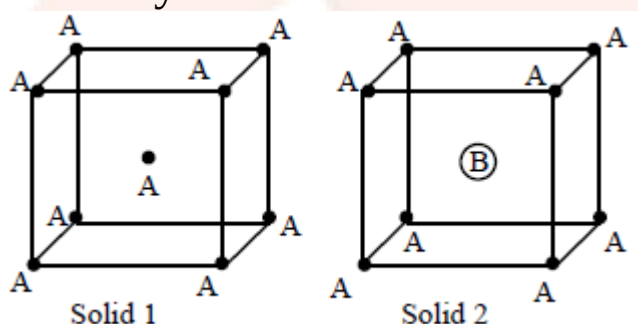
17. The maximum prescribed concentration of copper in drinking water is:

- A. 5 ppm B. 0.05 ppm
C. 3 ppm D. 0.5 ppm

Ans. C

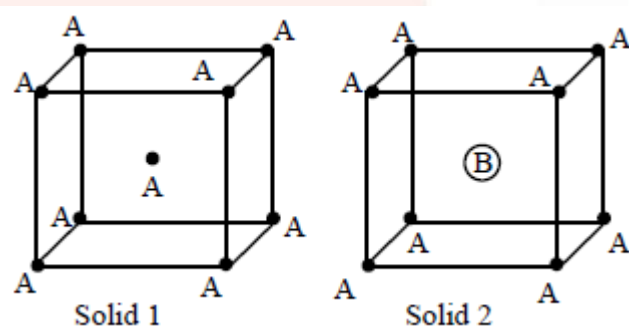
The prescribed conc. of Cu in drinking water is 3ppm.

- 18.** Consider the bcc unit cells of the solids 1 and 2 with the position of atoms as shown below. The radius of atom B is twice that of atom A. The unit cell edge length is 50% more in solid 2 than in 1. What is the approximate packing efficiency in solid 2?



- A. 45% B. 65%
C. 90% D. 75%

Ans. C



Given that the radius of atom B is twice that of atom A and The unit cell edge length is 50% more in solid 2 than in 1. So,

$$r_B = 2r_A \quad a_2 = 1.5 a_1$$

$$4r_A = r\sqrt{3} a_1, a_1 = \frac{4r_A}{\sqrt{3}}$$

$$a_2 = 1.5 a_1$$

$$= \frac{3}{2} \frac{4r_A}{\sqrt{3}}$$

$$a_2 = 2\sqrt{3} r_A$$

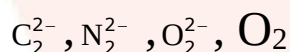
$$PE_2 = \frac{\left(\frac{4}{3} \pi r_A^3 \times 1\right) + \left(\frac{4}{3} \pi r_B^3 \times 1\right)}{a_2^3}$$

$$= \frac{\frac{4}{3} \pi r_A^3 + \frac{4}{3} \pi (2r_A)^3}{(2\sqrt{3} r_A)^3}$$

$$= \frac{\frac{4}{3} \pi r_A^3 \times 9}{8 \times 3\sqrt{3} r_A^3} = \frac{\pi}{2\sqrt{3}} = 90.64\%$$

$$= 90\%$$

19. Among the following molecules/ions,



Which one is diamagnetic and has the shortest bond length?



Ans. C

Paramagnetic compounds (and atoms) are those which are attracted to magnetic fields while diamagnetic compounds (and atoms) are those which repelled from magnetic fields.

Paramagnetic compounds have unpaired electrons while in diamagnetic compounds the electrons all have paired spins.

So, O_2 , N_2^{-2} = paramagnetic

C_2^{-2} and O_2^{-2} = diamagnetic

C_2^{-2} has B.O. = 3

∴ diamagnetic & shortest B.L.

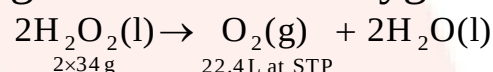
20. The strength of 11.2 volume solution of H_2O_2 is: [Given that molar mass of H=1g mol⁻¹ and O=16 g mol⁻¹]

A. 3.4% B. 34%

C. 13.6% D. 1.7%

Ans. A

11.2 volumes of H_2O_2 means that 1 mL of this H_2O_2 will give 11.2 mL of oxygen at STP.



11.2 vol. H_2O_2

m = ?

Vol. strength = 11.2 M

11.2 = 11.2 m

m = 1

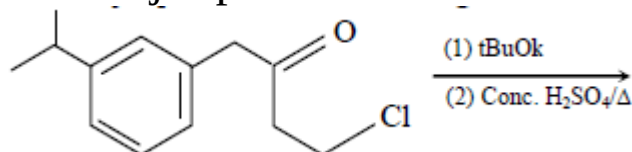
1 mole H_2O_2 present in 1L solution

34 gm in 1000 gm solution

$$\% \frac{w}{w} = \frac{(g_m)_{\text{solute}}}{(g_m)_{\text{solution}}} \times 100$$

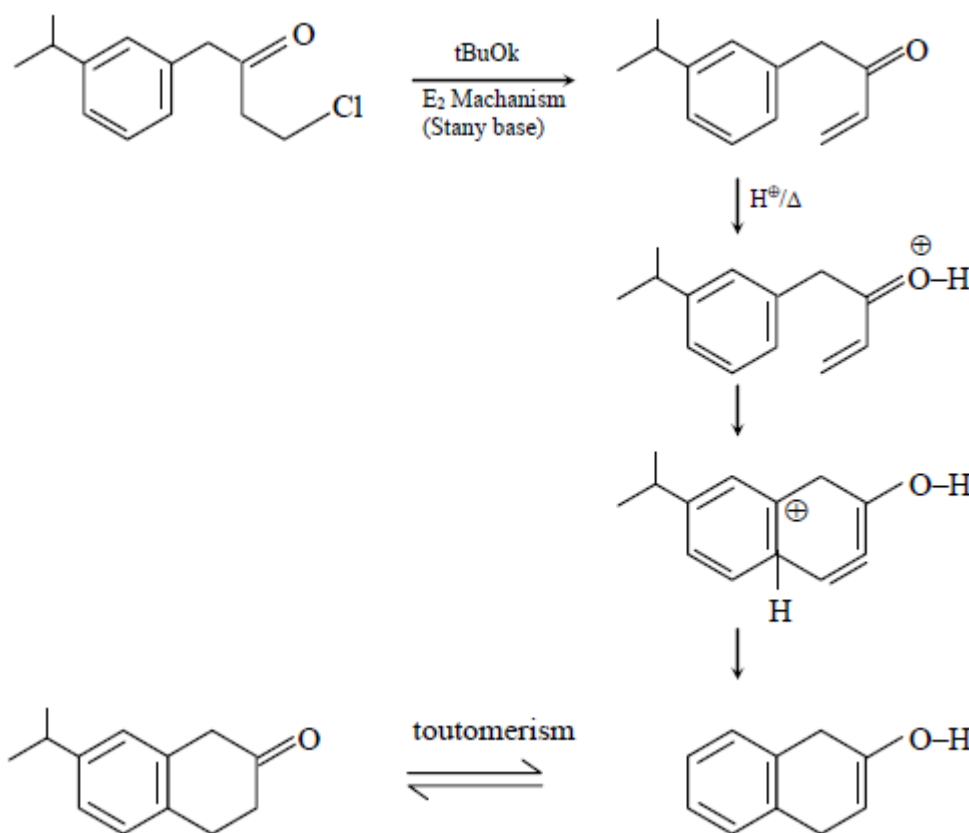
$$= \frac{34}{1000} \times 100 = 3.4\%$$

21. The major product of the following reaction is:



- A.
- B.
- C.
- D.

Ans. A

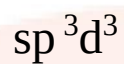
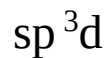


22. The ion that has sp^3d^2 hybridization for the central atom is:

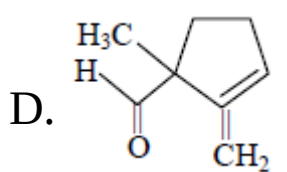
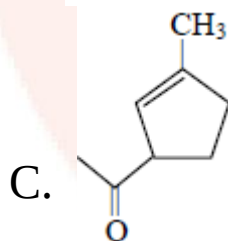
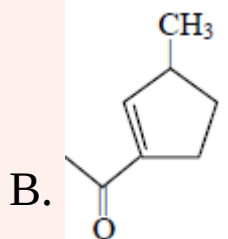
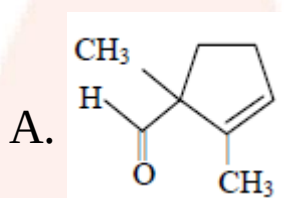
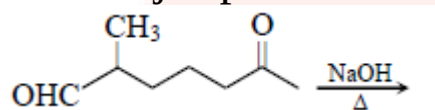
- A. $[\text{ICl}_2]^-$ B. $[\text{ICl}_4]^-$
C. $[\text{BrF}_2]^-$ D. $[\text{IF}_6]^-$

Ans. B

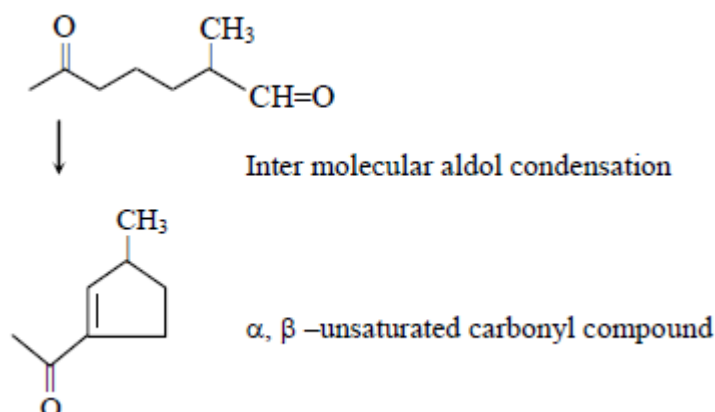
Species	Hybridisation
$[\text{ICl}_2]^-$	sp^3d
$[\text{ICl}_4]^-$	sp^3d^2



23. The major product obtained in the following reaction is:



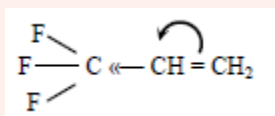
Ans. B



24. Which one of the following alkenes when treated with HCl yields majorly an anti Markovnikov product?

- A. $\text{F}_3\text{C}-\text{CH}=\text{CH}_2$
- B. $\text{Cl}-\text{CH}=\text{CH}_2$
- C. $\text{CH}_3\text{O}-\text{CH}=\text{CH}_2$
- D. $\text{H}_2\text{N}-\text{CH}=\text{CH}_2$

Ans. A



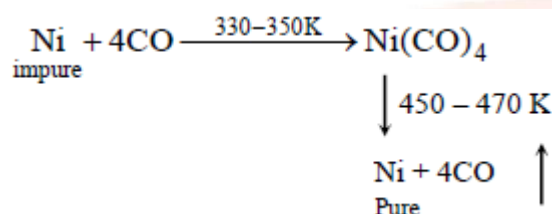
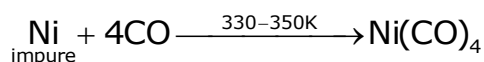
– $\text{CF}_3 \rightarrow -\text{H}$ effect and most e^- withdrawing group.

25. The Mond process is used for the:

- A. extraction of Zn
- B. purification of Zr and Ti
- C. extraction of Mo
- D. purification of Ni

Ans. D

Mond process is used in the purification of Ni. It is a vapour phase refining process.



26. The compound that inhibits the growth of tumors is:

- A. *cis*-[Pd(Cl)₂(NH₃)₂]
- B. *cis*-[Pt(Cl)₂(NH₃)₂]
- C. *trans*-[Pd(Cl)₂(NH₃)₂]
- D. *trans*-[Pt(Cl)₂(NH₃)₂]

Ans. B

cis-[Pt(Cl)₂(NH₃)₂] is known as *cis*-platin. It is a σ -bonded organo-metallic compound and is used as an anti-tumor in the treatment of cancer.

27. The percentage composition of carbon by mole in methane is:

- A. 75% B. 20%
- C. 25% D. 80%

Ans. B

In CH₄

Mole of carbon = 1

mole of hydrogen = 4

% composition of C by mole in CH₄

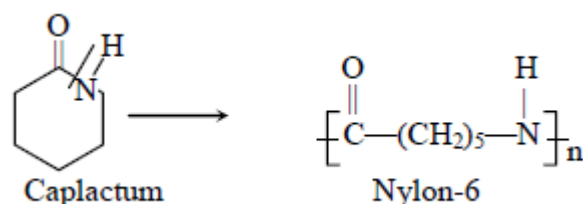
$$\begin{aligned}\%C &= \frac{1}{5} \times 100 \\ &= 20\%\end{aligned}$$

28. The structure of Nylon-6 is:

- A. $\left[(\text{CH}_2)_6 - \overset{\text{O}}{\parallel} \text{C} - \overset{\text{H}}{\underset{|}{\text{N}}} \right]_n$
 B. $\left[\overset{\text{O}}{\parallel} \text{C} - (\text{CH}_2)_6 - \overset{\text{H}}{\underset{|}{\text{N}}} \right]_n$
 C. $\left[(\text{CH}_2)_4 - \overset{\text{O}}{\parallel} \text{C} - \overset{\text{H}}{\underset{|}{\text{N}}} \right]_n$
 D. $\left[\overset{\text{O}}{\parallel} \text{C} - (\text{CH}_2)_5 - \overset{\text{H}}{\underset{|}{\text{N}}} \right]_n$

Ans. D

Nylon-6 is prepared by ring opening polymerization of caprolactum. It is heated about 533K in an inert atmospheric nitrogen about 4-5 hrs.



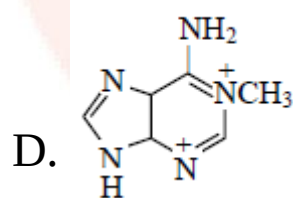
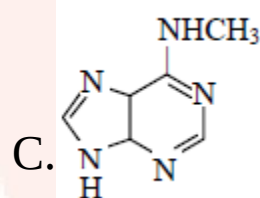
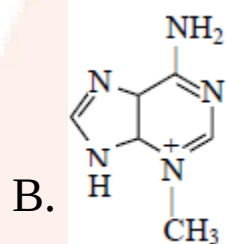
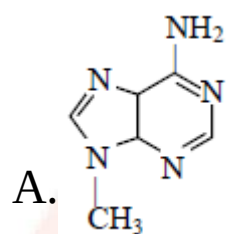
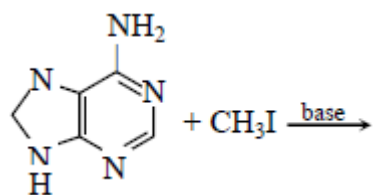
29. The IUPAC symbol for the element with atomic number 119 would be:

- A. uue B. uun
 C. une D. unh

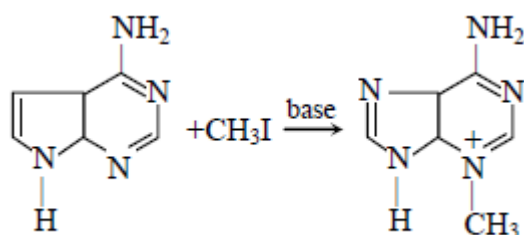
Ans. A

∴ IUPAC symbol for 119 is uue

30. The major product in the following reaction is:



Ans. A



Mathematics

1. If the lengths of the sides of a triangle are in A.P. and the greatest angle is double the smallest, then a ratio of lengths of the sides of this triangle is:
- A. 5:9:13 B. 5:6:7
C. 3:4:5 D. 4:5:6

Ans. D

Given that

$a < b < c$ are in A.P.

$\angle C = 2 \angle A$ (Given)

$$\Rightarrow \sin C = \sin 2A$$

$$\Rightarrow \sin C = 2 \sin A \cdot \cos A$$

$$\Rightarrow \frac{\sin C}{\sin A} = 2 \cos A$$

$$\Rightarrow \frac{c}{a} = 2 \frac{b^2 + c^2 - a^2}{2bc}$$

put $a = b - \lambda$, $c = b + \lambda$, $\lambda > 0$

$$\Rightarrow \lambda = \frac{b}{5}$$

$$\Rightarrow a = b - \frac{b}{5} = \frac{4}{5}b, c = b + \frac{b}{5} = \frac{6b}{5}$$

$$\Rightarrow \text{required ratio} = 4 : 5 : 6$$

2. If the eccentricity of the standard hyperbola passing through the point (4, 6) is 2, then the equation of the tangent to the hyperbola at (4, 6) is:

A. $2x - 3y + 10 = 0$ B. $3x - 2y = 0$

C. $2x - y - 2 = 0$ D. $x - 2y + 8 = 0$

Ans. C

Let equation of hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Hyperbola passing through the point (4, 6)

$$\frac{16}{a^2} - \frac{36}{b^2} = 1 \quad \dots(1)$$

$$a^2 = 1 + \frac{b^2}{a^2}$$

$$4 = 1 + \frac{b^2}{a^2}$$

$$b^2 = 3a^2 \quad \dots(ii)$$

from (i) & (ii)

$$a^2 = 4$$

$$b^2 = 12$$

$$\frac{x^2}{4} - \frac{y^2}{12} = 1$$

The equation of the tangent to the hyperbola at (4, 6) is

$$\frac{4x}{4} - \frac{6y}{12} = 1$$

$$x - \frac{y}{2} = 1$$

$$2x - y - 2 = 0$$

3. The sum $\sum_{k=1}^{20} k \frac{1}{2^k}$ is equal to:

A. $1 - \frac{11}{2^{20}}$ B. $2 - \frac{21}{2^{20}}$

C. $2 - \frac{11}{2^{19}}$ D. $2 - \frac{3}{2^{17}}$

Ans. C

It is given that given that

$$S = \sum_{k=1}^{20} \frac{k}{2^k}$$

$$S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{20}{2^{20}} \quad \dots(i)$$

$$\frac{1}{2}S = \frac{1}{2^2} + \frac{2}{2^2} + \dots + \frac{20}{2^{21}} \quad \dots(ii)$$

Now solving the equation (i) and (ii) and subtracting equation (ii) from (i), we get,

$$\frac{1}{2}S - \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^{20}} \right) - \frac{20}{2^{21}}$$

$$\frac{1}{2}S = \frac{\frac{1}{2} \left(1 - \frac{1}{2^{20}} \right)}{1 - \frac{1}{2}} - \frac{20}{2^{21}}$$

$$\frac{1}{2}S = 1 - \frac{1}{2^{20}} - \frac{20}{2^{21}}$$

$$\frac{1}{2}S = 1 - \frac{1}{2^{20}} - \frac{10}{2^{20}}$$

$$\frac{1}{2}S = 1 - \frac{11}{2^{20}}$$

$$S = 2 - \frac{11}{2^{19}}$$

Thus, the sum $\sum_{k=1}^{20} k \frac{1}{2^k}$ is equal to $2 - \frac{11}{2^{19}}$.

4. A student scores the following marks in five tests: 45, 54, 41, 57, 43. His score is not known for the sixth test. If the mean score is 48 in the six tests, then the standard deviation of the marks in six tests is:

A. $\frac{100}{\sqrt{3}}$

B. $\frac{100}{3}$

C. $\frac{10}{\sqrt{3}}$

D. $\frac{10}{3}$

Ans. C

Let the marks in sixth tests is 'x' so mean

$$\frac{45 + 54 + 41 + 57 + 43 + x}{6} = 48$$

$$\frac{240 + x}{6} = 48$$

$$x = 48$$

Now, standard deviation of these marks

$$\sigma^2 = \left(\frac{1}{6} (45^2 + 54^2 + 41^2 + 51^2 + 43^2 + 48^2) \right) - (48)^2$$

$$\sigma^2 = \frac{1}{6} \times (14024) - (48)^2$$

$$\sigma^2 = \frac{14024 - ((48)^2 \times 6)}{6}$$

$$\sigma^2 = \frac{14024 - 13824}{6}$$

$$\sigma^2 = \frac{200}{6} = \frac{100}{3}$$

$$\sigma^2 = \frac{10}{\sqrt{3}}$$

5. If a point R(4, y, 2) lies on the line segment joining the points P(2, — 3, 4) and Q(8, 0, 10), then the distance of R from the origin is:

- A. $2\sqrt{14}$ B. $2\sqrt{21}$
C. $\sqrt{53}$ D. 6

Ans. A

The line equation segment joining the points P(2, — 3, 4) and Q(8, 0, 10) is given by

$$\frac{x-2}{6} = \frac{y+3}{3} = \frac{z-4}{6} = \lambda$$

Let point R ($6\lambda + 2$, $3\lambda - 3$, $6\lambda + 4$)

given that $6\lambda + 2 = 4$

point R (4, -2, 6)

Distance between point R and origin $= \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$

6. The number of Integral values of m for which the equation $(1 + m^2)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$ has no real root is :
 A.1 B. infinitely many
 C.2 D. 3

Ans. B

The equation $(1 + m^2)x^2 - 2(1 + 3m)x + (1 + 8m) = 0$

Equation has no real solution, $D < 0$

$$4(1 + 3m)^2 - 4 \times (1 + m^2)(1 + 8m) < 0$$

$$1 + 9m^2 + 6m - (1 + 8m + m^2 + 8m^3) < 0$$

$$1 + 9m^2 + 6m - 1 - 8m - m^2 - 8m^3 < 0$$

$$-8m^3 + 8m^2 - 2m < 0$$

$$8m^3 - 8m^2 + 2m > 0$$

$$m(4m^2 - 4m + 2) > 0$$

$$m[(2m - 1)^2 + 1] > 0$$

$$m > 0$$

$\Rightarrow$ Number of integral values of m are infinitely many

7. Let the number 2, b, c be in an A.P. and

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & b & c \\ 4 & b^2 & c^2 \end{bmatrix}. \text{ If } \det(A) \in [2, 16], \text{ then c lies in the interval:}$$

$$A. [3, 2 + 2^{3/4}] \quad B. (2, 3)$$

C. [4,6] D. $(2+2^{3/4}, 4)$

Ans. C

Given, matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & b & c \\ 4 & b^2 & c^2 \end{bmatrix}$, so

$$\det(A) = \begin{vmatrix} 1 & 1 & 1 \\ 2 & b & c \\ 4 & b^2 & c^2 \end{vmatrix}$$

on applying, $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$\text{We get } \det(A) = \begin{vmatrix} 1 & 0 & 0 \\ 2 & b-2 & c-2 \\ 4 & b^2-4 & c^2-4 \end{vmatrix}$$

$$= \begin{vmatrix} b-2 & c-2 \\ b^2-4 & c^2-4 \end{vmatrix}$$

$$= \begin{vmatrix} b-2 & c-2 \\ (b-2)(b+2) & (c-2)(c+2) \end{vmatrix}$$

[Taking common (b-2) from C_1 and (c-2) from C_2]

$$|A| = (2-b)(b-c)(c-2)$$

2, b, c, are in AP.

$$b = \frac{2+c}{2}$$

$$\text{Det}(A) = \frac{1}{4}(c-2)^3$$

$$2 \leq \frac{1}{4}(c-2)^3 \leq 16$$

$$8 \leq (c-2)^3 \leq 64$$

$$2 \leq c-2 \leq 4$$

$$4 \leq c \leq 6$$

8. The minimum number of times one has to toss a fair coin so that the probability of observing at least one head is at least 90% is:

A. 5 B. 4
C. 3 D. 2

Ans. B

The required probability of observing atleast one head = $1 - P(\text{no head})$

$$P(\text{at least one heads}) = 1 - [(\text{no one heads})] \geq \frac{90}{100}$$

$$1 - \left(\frac{1}{2}\right)^n \geq \frac{9}{100}$$

$$\left(\frac{1}{2}\right)^n \leq \frac{10}{100} \Rightarrow 2^n \geq 10 \Rightarrow n \geq 4$$

The minimum number of times one has to toss a fair coin so that the probability of observing at least one head is at least 90% is 4

9. The tangent to the parabola $y^2=4x$ at the point where it intersects the circle $x^2+y^2=5$ in the first quadrant, passes through the point:

A. $\left(\frac{3}{4}, \frac{7}{4}\right)$ B. $\left(-\frac{1}{4}, \frac{1}{2}\right)$
 C. $\left(-\frac{1}{3}, \frac{4}{3}\right)$ D. $\left(\frac{1}{4}, \frac{3}{4}\right)$

Ans. A

Given equations of the parabola $y^2 = 4x$... (i)

and circle $x^2 + y^2 = 5$... (ii)

So, for point of intersection of curves (i) and (ii), put $y^2 = 4x$ in Eq. (ii) we get

$$x^2 + 4x - 5 = 0$$

$$(x + 5)(x - 1) = 0$$

$$x = -5 \text{ or } x = 1$$

not possible $y = \pm 2$

Point in IQ (1, 2)

Tangent at (1, 2)

$$2y = 4\left(\frac{x+1}{2}\right)$$

$$y = x + 1$$

point $\left(\frac{3}{4}, \frac{7}{4}\right)$ lies on tangent

10. If the system of linear equations

$$x - 2y + kz = 1$$

$$2x + y + z = 2$$

$$3x - y - kz = 3$$

has a solution $(x, y, z), z \neq 0$, then (x, y) lies on the straight line whose equation is :

A. $4x - 3y - 4 = 0$ B. $3x - 4y - 1 = 0$

C. $3x-4y-4=0$ D. $4x-3y-1=0$

Ans. A

Given system of linear equations

$x - 2y + kz = 1 \dots (i)$

$2x + y + z = 2 \dots (ii)$

$3x - y - kz = 3 \dots (iii)$

for locus of (x, y)

equation (i) + (iii)

$4x - 3y = 4$

$4x - 3y - 4 = 0$

This is the required equation of the straight line in which point (x, y) lies.

11. Which one of the following statements is not a tautology?

A. $(p \wedge q) \rightarrow p$

B. $(p \vee q) \rightarrow (p \vee (\sim q))$

C. $p \rightarrow (p \vee q)$

D. $(p \wedge q) \rightarrow (\sim p)q$

Ans. B

p	q	$p \vee q$	$\sim q$	$p \vee \sim q$	$(p \vee q) \rightarrow (p \vee \sim q)$
T	T	T	F	T	T
T	F	T	T	T	T
F	T	T	F	F	F
F	F	F	T	T	T

12. Let $f: [-1, 3] \rightarrow \mathbb{R}$ be defined as

$$F(x) = \begin{cases} |x| + [x], & -1 \leq x < 1 \\ x + |x|, & 1 \leq x < 2 \\ x + [x], & 2 \leq x < 3 \end{cases}$$

Where $[t]$ denotes the greatest integer less than or equal to t .
Then, f is discontinuous at:

- A. only one point
- B. only three points
- C. only two points
- D. four or more points

Ans. B

$$\begin{cases} |x| + [x] & -1 \leq x < 1 \\ x + |x| & 1 \leq x < 2 \\ x + [x] & 2 \leq x \leq 3 \end{cases}$$

$f \times n$ is discontinuous
at $x = 0, 1, 3$

13. If $\int \frac{dx}{x^3(1+x^6)^{2/3}} = x f(x) (1+x^6)^{1/3} + C$

Where, C is a constant of integration, then the function $f(x)$ is equal to:

- A. $-\frac{1}{2x^2}$
- B. $-\frac{1}{2x^3}$
- C. $-\frac{1}{6x^3}$
- D. $\frac{3}{x^2}$

Ans. B

It is given that

$$\begin{aligned}
 I &= \int \frac{dx}{x^3(1+2x^6)^{2/3}} \\
 &= \int \frac{dx}{x^3 \cdot x^4 \left(\frac{1}{x^6} + 1\right)^{2/3}} = \int \frac{dx}{x^7 \left(\frac{1}{x^6} + 1\right)^{2/3}} \\
 I &= \int \frac{x^{-7} dx}{(x^{-6} + 1)^{2/3}} \text{----- (i)}
 \end{aligned}$$

Now, putting $t = x^{-6} + 1$ and differentiating it, we get,

$$\begin{aligned}
 dt &= -6x^{-7} dx \\
 -\frac{1}{6} dt &= x^{-7} dx
 \end{aligned}$$

So, now putting the value of dx in equation (i), we get,

$$\begin{aligned}
 I &= -\frac{1}{6} \int t^{-2/3} dt \\
 I &= -\frac{1}{6} \times 3(t)^{1/3} + c \\
 I &= -\frac{1}{2} \left(\frac{1+x^6}{x^6}\right)^{1/3} + c \\
 I &= -\frac{1}{2} (1+x^6)^{1/3} + c
 \end{aligned}$$

So, the function $f(x)$ is equal to:

$$\begin{aligned}
 xf(x) &= \frac{1}{2x^2} \\
 f(x) &= -\frac{1}{2x^2}
 \end{aligned}$$

14. Let $\vec{a} = 3\hat{i} + 2\hat{j} + x\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$, for some real x . Then $|\vec{a} \times \vec{b}| = r$ is possible if:

A. $0 < r \leq \sqrt{\frac{3}{2}}$

B. $\sqrt{\frac{3}{2}} < r \leq 3\sqrt{\frac{3}{2}}$

C. $3\sqrt{\frac{3}{2}} < r \leq 5\sqrt{\frac{3}{2}}$

D. $r \geq 5\sqrt{\frac{3}{2}}$

Ans. D

Given vectors are $\vec{a} = 3\hat{i} + 2\hat{j} + x\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & x \\ 1 & -1 & 1 \end{vmatrix}$$

$$\vec{a} \times \vec{b} = \hat{i}(2+x) - \hat{j}(3-x) + \hat{k}(-5)$$

$$\vec{a} \times \vec{b} = (2+x)\hat{i} - (x-3)\hat{j} - 5\hat{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{(2+x)^2 + (x-3)^2 + 25}$$

$$= \sqrt{2x^2 - 2x + 38}$$

$$\geq \sqrt{\frac{75}{2}}$$

$$\geq 5\sqrt{\frac{3}{2}}$$

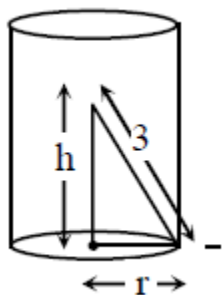
15. The height of a light circular cylinder of maximum volume inscribed in a sphere of radius 3 is:

A. $\frac{2}{3}\sqrt{3}$ B. $\sqrt{6}$

C. $\sqrt{3}$ D. $2\sqrt{3}$

Ans. D

Let a sphere of radius 3, which inscribed a right circular cylinder having radius r and height is h ,



$$h^2 + r^2 = 9$$

$$r^2 = 9 - h^2$$

$$V = \pi r^2(2h)$$

$$v = 2\pi h (9 - h^2)$$

$$v = 2\pi (9 - 4h^3)$$

$$\frac{dv}{dx} = 0 \quad h = \sqrt{3}$$

$$r = \sqrt{6}$$

$$V_{\max} = \pi r^2(2h)$$

$$= \pi 6(2\sqrt{3})$$

$$H = 2\sqrt{3} = 2(h)$$

16. If $f(1)=1$, $f'(1)=3$, then the derivative of $f(f(f(x))) + (f(x))^2$ at $x=1$ is:

A.15 B. 33

C.12 D. 9

Ans. B

Let $y = f(f(f(x))) + (f(x))^2$ at $x=1$

On differentiating both sides with respect to x we get

$f(f(f(x))) + (f(x))^2$ at $x = 1$

$$\begin{aligned}
 & f'(f(f(x))) \cdot f'(f(x)) \cdot f'(x) + 2f(x) f'(x) \\
 & f'(f(f(x))) \cdot f'(f(x)) \cdot f'(x) + 2f(x) f'(1) \\
 & f'(f(f(1))) \cdot f'(1) \cdot f'(1) + 2f'(1) \\
 & f'(f(1)) \cdot f'(1) \cdot f'(1) + 2f'(1) \\
 & 3 \times 3 \times 3 + (2 \times 3) f'(1) = 1 f'(1) = 3 \\
 & 27 + 6 = 33
 \end{aligned}$$

17. Suppose that the points (h, k) , $(1, 2)$ and $(-3, 4)$ lie on the line L_1 . If a line L_2 passing through the points (h, k) and $(4, 3)$ is perpendicular to L_1 , then k/h equals:

- A. $-\frac{1}{7}$ B. $\frac{1}{3}$
C. 3 D. 0

Ans. B

Equation of line passes through $(1, 2)$ and $(-3, 4)$

$$(y - 2) = \frac{4 - 2}{-3 - 1}(x - 1)$$

$$(y - 2) = -\frac{1}{2}(x - 1)$$

$$x + 2y = 5 \dots (i)$$

$\perp$ line $2x - y = \lambda \rightarrow$ passes through $(4, 3)$

$$2x - y = 5 \dots (2) \lambda = 5$$

Intersection point of line (i) & line (ii) is $(3, 1)$

$$\frac{k}{h} = \frac{1}{3}$$

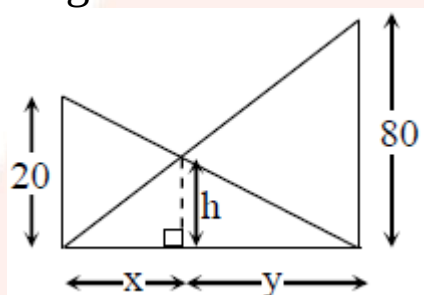
18. Two vertical poles of heights, 20 m and 80m apart on a horizontal plane. The height (in meters) of the point of

intersection of the lines joining the top of each pole to the foot of the other, from this horizontal plane is:

- A.15 B. 12
C.16 D. 18

Ans. C

Let s first pole having height 20 m and second pole having height 80 m.



$$\frac{h}{x} = \frac{80}{x+y} \quad \frac{h}{y} = \frac{20}{x+y}$$

$$h = \frac{80}{1+(y/x)} \dots (i) \quad h = \frac{20}{\left(\frac{x}{y}\right)+1}$$

$$h = \frac{80}{1+\frac{h}{20-h}} \quad \frac{x}{y} + 1 = \frac{20}{h}$$

$$h = 4(20-h) \quad \frac{x}{y} = \frac{20}{h} - 1$$

$$h = 80 - 4h \quad \frac{x}{y} = \frac{20-h}{h}$$

$$5h = 80 \quad \frac{x}{y} = \frac{h}{20-h}$$

$$h = 16 \quad \text{put in } \dots (i)$$

19. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function satisfying $f'(3) + f'(2) = 0$.

Then $\lim_{x \rightarrow 0} \left(\frac{1 + f(3+x) - f(3)}{1 + f(3+x) - f(2)} \right)^{\frac{1}{x}}$

Is equal to:

- A. e^2 B. 1
C. e^{-1} D. e

Ans. B

$$\begin{aligned} & \lim_{x \rightarrow 0} \left(\frac{1 + f(3+x) - f(3)}{1 + f(2-x) - f(2)} \right)^{\frac{1}{x}} \quad \text{Form (1)}^\infty \\ &= e^{\lim_{x \rightarrow 0} \frac{1}{x} \left[\frac{1 + f(3+x) - f(3)}{1 + f(2-x) - f(2)} - 1 \right]} \\ &= e^{\lim_{x \rightarrow 0} \frac{1}{x} \left[\frac{f(3+x) - f(3) - f(2-x) + f(2)}{1 + f(2-x) - f(2)} \right]} \quad \text{Form } \left(\frac{0}{0} \right) \\ &= e^{\lim_{x \rightarrow 0} \left(\frac{f(3+x) - f(3) - f(2-x) + f(2)}{x} \right) \times \left(\frac{1}{1} \right)} \\ &= e^{\lim_{x \rightarrow 0} \frac{f'(3+x) + f'(2-x)}{1}} \\ &= e^{f'(3) + f'(2)} = e^0 = 1 \\ &= e \end{aligned}$$

20. If three distinct numbers a, b, c are in G.P. and the equations $ax^2 + 2bx + c = 0$ and $dx^2 + 2ex + f = 0$ have a common root, then which one of the following statements is correct?

- A. d, e, f are in G.P.
B. $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.
C. d, e, f are in A.P.
D. $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.

Ans. D

Given three distinct numbers a, b and c are in GP

$$\therefore b^2 = ac \dots(i)$$

and the given quadratic equations

$$ax^2 + 2bx + c = 0 \dots(ii)$$

$$dx^2 + 2ex + f = 0 \dots(iii)$$

$$(af - cd)^2 = (2ae - 2bd)(2bf - 2ec)$$

$$a^2f^2 + c^2d^2 - 2a + cd = 4aebf - 4ae^2c - 4b^2df + 4bdec$$

$$a^2f^2 + c^2d^2 + 4b^2e^2 + 2afcd - 4aebf - 4bdec = 0$$

$$(af + cd - 2be)^2 = 0$$

$$af + cd = 2be$$

$$\frac{af}{b^2} + \frac{cd}{b^2} = \frac{2be}{b^2}$$

$$\frac{af}{ac} + \frac{cd}{ac} = 2\left(\frac{e}{b}\right)$$

$$\frac{f}{c} + \frac{d}{a} = 2\left(\frac{e}{b}\right)$$

$$\frac{d}{a}, \frac{e}{b}, \frac{f}{c} \text{ are in AP.}$$

21. The tangent and the normal lines at the point $(\sqrt{3}, 1)$ to the circle $x^2 + y^2 = 4$ and the x-axis form a triangle. The area of this triangle (in square units) is:

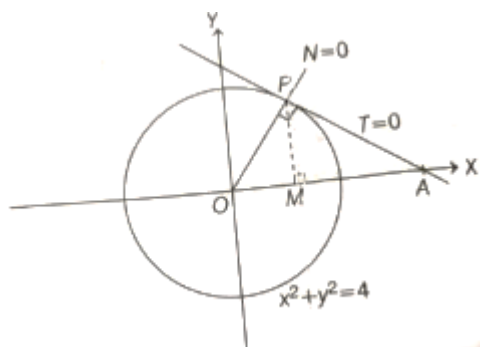
A. $\frac{2}{\sqrt{3}}$

B. $\frac{1}{\sqrt{3}}$

C. $\frac{1}{3}$

D. $\frac{4}{\sqrt{3}}$

Ans. A



So, equation of tangent ($T = 0$) is

$$\sqrt{3}x + y = 4$$

For point A, put $y = 0$, we get

$$x = \frac{4}{\sqrt{3}}$$

$\therefore$ Area of required $\Delta OPA = \frac{1}{2}(OA)(PM)$

$$x = \frac{4}{\sqrt{3}}$$

[$\therefore PM = y$ -coordinate of P]

$$= \frac{2}{\sqrt{3}} \text{ sq unit .}$$

22. If the fourth term in the binomial expansion of $\left(\sqrt{\frac{1}{x^{1+\log_{10} x}} + x^{\frac{1}{2}}}\right)^6$ is equal to 200, and $x > 1$, then the value of x is:

A. 10^4 B. 10

C. 10^3 D. 100

Ans. B

It is given that binomial is $\left(\sqrt{\frac{1}{x^{1+\log_{10} x}} + x^{\frac{1}{2}}}\right)^6$

Since the fourth term in the expansion is 200,
So

$$\left(\left(\frac{1}{x^{1+\log_{10}^x}}\right)^{\frac{1}{2}} + x^{\frac{1}{12}}\right)^6 \text{ and } T_4 = 200$$

$$T_4 = {}^6C_3 (x)^{-\frac{3}{2}(1+\log_{10}^x)} \cdot \left(x^{\frac{1}{12}}\right)^3 = 200$$

$$20 \cdot x^{-\frac{3}{2}(1+\log_{10}^x)} + \frac{1}{4} = 200$$

$$x^{-\frac{3}{2}(1+\log_{10}^x) + \frac{1}{4}} = 10$$

Taking, log on both side, we get,

$$\left(-\frac{3}{2}(1 + \log_{10}^x) + \frac{1}{4}\right) \log_{10}^x = 1$$

Now, let $\log_{10}^x = t$, we get,

$$-6(t^2 + t) + \frac{t}{4} = 1$$

$$-6(t^2 + t) + t = 4$$

$$-6t^2 - 6t + t = 4$$

$$6t^2 + 5t + 4 = 0$$

$\Delta < 0$ and roots are imaginary.

(This is a bonus question)

23. Let $f(x) = \int_0^x g(t) dt$, where g is a non-zero even function. If

$f(x+5) = g(x)$, then $\int_0^x f(t) dt$ equal:

A. $\int_5^{x+5} g(t) dt$ B. $\int_{x+5}^5 g(t) dt$

C. $5 \int_{x+5}^5 g(t) dt$ D. $2 \int_5^{x+5} g(t) dt$

Ans. B

Given $f(x) = \int_0^x g(t) dt$

On replacing x by $(-x)$ we get

$$f(-x) = \int_0^{-x} g(t) dt$$

Now, put $t = -u$, so

$$f(-x) = -\int_0^x g(-u) du = -\int_0^x g(u) du = -f(x)$$

[$\because g$ is an even function]

$$f(0) = \int_0^0 g(t) dt = 0$$

$f(0) = 0$ $f(x)$ is odd function

give that $g(x)$ is even function

$$f(x+5) = f(-x+5) = g(x) = g(-x)$$

$$I = \int_0^x f(t) dt$$

$$z = t + 5$$

$$I = \int_5^{x+5} f(3 - 5) dz$$

$$I = \int_5^{x+5} f(-(f - 3)) dz$$

$$I = \int_5^{x+5} f(5 - 3) dz$$

$$I = \int_{x+5}^5 f(5 - z) dz$$

$$I = \int_{x+5}^5 g(z) dz$$

$$I = \int_{x+5}^5 g(t) dt$$

24. Given that the slope of the tangent to a curve $y=y(x)$ at any Point (x,y) is $\frac{2y}{x^2}$ If the curve passes through the center of the circle $x^2+y^2-2x-2y=0$, then its equation is:

A. $x^2 \log_e |y| = -2(x-1)$

B. $x \log_e |y| = -2(x-1)$

C. $x \log_e |y| = x-1$

D. $x \log_e |y| = 2(x-1)$

Ans. D

slope of the tangent to a curve $y=y(x)$ at any Point (x,y) is given

$$\frac{dy}{dx} = \frac{2y}{x^2}$$

$$\frac{dy}{y} = \frac{2}{x^2} dx$$

Integrating both sides

$$\log_e |y| = -\frac{2}{x} + c$$

process through (1, 1)

$$0 = -2 + c \quad c = 2$$

$$\log_e |y| = -\frac{2}{x} + 2$$

$$x \log_e |y| = -2 + 2x$$

$$x \log_e |y| = 2(x - 1)$$

25. If $z = \frac{\sqrt{3}}{2} + \frac{i}{2}$ ($i = \sqrt{-1}$), then $(1 + iz + z^5 + iz^8)^9$ is equal to:

- A. 1 B. 0
C. -1 D. $(-1 + 2i)^9$

Ans. C

$$z = \frac{\sqrt{3}}{2} + \frac{i}{2} = -i \left(-\frac{1}{2} + \frac{\sqrt{3}}{2} i \right) = -i\omega$$

$$(1 + iz + z^5 + iz^8)^9 = (1 + \omega - i\omega^2 + i\omega^2)^9 = (1 + \omega^2)^9 = (-\omega^2)^9 = (-\omega^{18}) = -1$$

26. Let $f(x) = a^x$ ($a > 0$) be written as $f(x) = f_1(x) + f_2(x)$, where $f_1(x)$ is an even function and $f_2(x)$ is an odd function. Then $f_1(x+y) + f_1(x-y)$ equals:

- A. $2f_1(x+y)f_2(x-y)$
B. $2f_1(x)f_2(y)$

C. $2f_1(x+y)f_1(x-y)$

D. $2f_1(x)f_1(y)$

Ans. D

Given function $f(x) = a^x$ ($a > 0$) is written as sum of an even and odd function $f_1(x)$ and $f_2(x)$ respectively.

Clearly, $f_1(x) = \frac{a^x + a^{-x}}{2}$ and $f_2(x) = \frac{a^x - a^{-x}}{2}$

$$f(x) = \left(\frac{f(x) + f(-x)}{2} \right) + \left(\frac{f(x) - f(-x)}{2} \right)$$

↓
even

↓
odd

$$f_1(x) = \frac{f(x) + f(-x)}{2}$$

$$f_1(x+y) + f_1(x-y) = \frac{a^{x+y} + a^{-x-y}}{2} + \frac{a^{x-y} + a^{y-x}}{2}$$

$$= \frac{1}{2} [a^x (a^y + a^{-y}) + a^{-x} (a^{-x} + a^y)]$$

$$= \frac{1}{2} (a^x + a^{-x}) (a^y + a^{-y})$$

$$= \frac{1}{2} \left(\frac{a^x + a^{-x}}{2} \right) \left(\frac{a^y + a^{-y}}{2} \right)$$

$$= 2f_1(x) f_1(y)$$

27. Let $S(\alpha) = \{(x, y) : y^2 \leq x, 0 \leq x \leq \alpha\}$ and $A(\alpha)$ is the area of the region $S(\alpha)$. If for a $\lambda, 0 < \lambda < 4$, $A(\lambda) : A(4) = 2 : 5$, then λ equals:

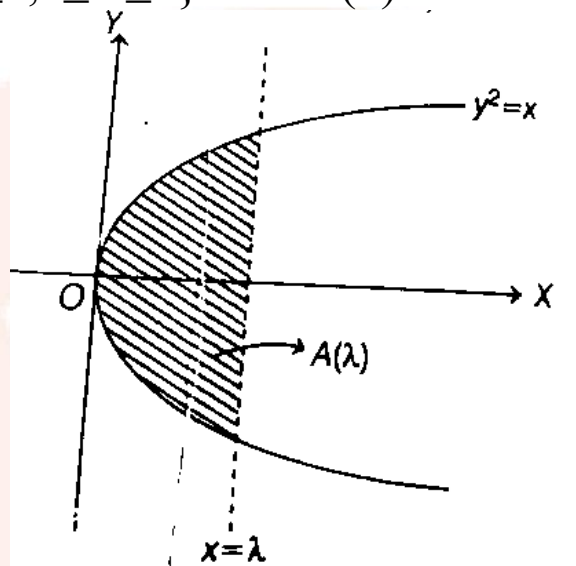
A. $2 \left(\frac{2}{5} \right)^{\frac{1}{3}}$ B. $4 \left(\frac{2}{5} \right)^{\frac{1}{3}}$

C. $4 \left(\frac{4}{25} \right)^{\frac{1}{3}}$ D. $2 \left(\frac{4}{25} \right)^{\frac{1}{3}}$

Ans. C

Given,

$S(\alpha) = \{(x,y): y^2 \leq x, 0 \leq x \leq \alpha\}$ and $A(\alpha)$ is area of the region $S(\alpha)$



$$\frac{A(\lambda)}{A(4)} = \frac{2}{5}$$

$$\frac{\int_0^{\lambda} \sqrt{x} \, dx}{\int_0^4 \sqrt{x} \, dx} = \frac{2}{5}$$

$$\frac{\lambda^{3/2}}{4^{3/2}} = \frac{2}{5}$$

$$\lambda^{3/2} = \frac{2}{5} \times 8$$

$$\lambda = \left(\frac{16}{5}\right)^{2/3}$$

$$\lambda = 4\left(\frac{2}{5}\right)^{2/3}$$

$$\lambda = 4\left(\frac{4}{25}\right)^{1/3}$$

28. In an ellipse, with center at the origin, if the difference of the lengths of major axis and minor axis is 10 and one of the foci is at $(0, 5\sqrt{3})$, then the length of its latus rectum is:

- A. 10 B. 8
C. 5 D. 6

Ans. C3

Given that $2b - 2a = 10$

$b - a = 5 \dots (i)$

given that

One of the focus of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is on y-axis $(0, 5\sqrt{3})$

$$2be = 10\sqrt{3}$$

$$be = 5\sqrt{3}$$

$$(b^2 - a^2) = 75$$

$$(b - a)(b + a) = 75$$

$$5(b + a) = 75$$

$$b + a = 15 \dots (ii)$$

from equation (i) & equation (2)

$$b = 10$$

$$a = 5$$

length of lotus rectum

$$= \frac{2a^2}{b}$$

$$= \frac{2 \times 25}{10} = 5$$

29. The vector equation of the plane through the line of intersection of the planes $x+y+z=1$ and $2x+3y+4z=5$ which is perpendicular to the plane $x-y+z=0$ is:

- A. $\vec{r} \cdot (\hat{i} - \hat{k}) + 2 = 0$
- B. $\vec{r} \cdot (\hat{i} - \hat{k}) - 2 = 0$
- C. $\vec{r} \times (\hat{i} - \hat{k}) + 2 = 0$
- D. $\vec{r} \times (\hat{i} - \hat{k}) + 2 = 0$

Ans. A

Since, equation of planes passes through the line of intersection of the planes

$$x + y + z = 1 \text{ and } 2x + 3y + 4z = 5$$

Now, the equation of required plane

$$(x + y + z - 1) + \lambda (2x + 3y + 4z - 5) = 0$$

$$x(1 + 2\lambda) + y(1 + 3\lambda) + z(1 + 4\lambda) - 5\lambda - 1 = 0 \dots (i)$$

$$x - y + 3z = 0 \dots (ii)$$

Plane (1) & (2) are perpendicular to each other

$$(1)(1 + 2\lambda) + (-1)(1 + 3\lambda) + (1)(1 + 4\lambda) = 0$$

$$1 + 2\lambda - 1 - 3\lambda + 1 + 4\lambda = 0$$

$$3\lambda = -1$$

$$\lambda = -\frac{1}{3}$$

put in equation (i)

$$\frac{x}{3} - \frac{z}{3} + \frac{2}{3} = 0$$

$$x - z + 2 = 0$$

$$\vec{r} \cdot (\hat{i} - \hat{k}) = -2$$

30. The number of four-digit numbers strictly greater than 4321 that can be formed using the digits 0, 1, 2, 3, 4, 5 (repetition of digits is allowed) is :

A. 306 B. 310

C. 360 D. 288

Ans. B

Following are the cases in which the 4-digit numbers strictly greater than 4321 can be formed using digits 0, 1, 2, 3, 4, 5 (repetition of digits is allowed)

Given digits are 0, 1, 2, 3, 4, 5

requires four number greater than 4321

$$\begin{array}{|c|c|c|c|} \hline 5 & & & \\ \hline \end{array} \begin{array}{c} \downarrow \\ 1 \end{array} \begin{array}{c} \downarrow \\ 6 \end{array} \begin{array}{c} \downarrow \\ 6 \end{array} \begin{array}{c} \downarrow \\ 6 \end{array} = 216$$

$$\begin{array}{|c|c|c|c|} \hline 4 & & & \\ \hline \end{array} \begin{array}{c} \downarrow \\ 2 \end{array} \begin{array}{c} \downarrow \\ 6 \end{array} \begin{array}{c} \downarrow \\ 6 \end{array} = 72$$

$$\begin{array}{|c|c|c|c|} \hline 4 & 3 & & \\ \hline \end{array} \begin{array}{c} \downarrow \\ 4 \end{array} \begin{array}{c} \downarrow \\ 6 \end{array} = 24$$

total case = 22

{subtract two case 4320 & 4321}

required total numbers = 216 + 72 + 22

$$= 310$$

